

2026-05-07-p2-f27-aider-voice-repomap

P2-F27 — Aider Voice Input + Repo-Map Implementation Plan

Programme position: F27 is the **seventh** Phase 2 feature (after F21-F26).

Goal: Ship real voice input (speech-to-text) and tree-sitter repository mapping. 4 tools + /aider slash. New packages: `internal/voice/`, extends `internal/repomap/`.

Spec: `docs/superpowers/specs/2026-05-07-p2-f27-aider-voice-repomap-design.md` **Q1-Q5:** Aider / Core / Hybrid Whisper / Tree-sitter / 4 tools + slash

Zero new external deps — `go-tree-sitter` already indirect, `whisper.cpp` is CLI binary not Go dep.

Task list

- ☐ P2-F27-T01 — bootstrap F27 evidence + advance PROGRESS + CONTINUATION
 - ☐ P2-F27-T02 — `internal/voice/types.go` + `recorder.go`: VoiceRecorder with arecord/sox (TDD)
 - ☐ P2-F27-T03 — `internal/voice/transcriber.go`: Whisper API + `whisper.cpp` (TDD)
 - ☐ P2-F27-T04 — `internal/voice/voice_tools.go`: `voice_start/stop/transcribe` (TDD)
 - ☐ P2-F27-T05 — `internal/repomap/mapper.go` + `cache.go`: tree-sitter AST map (TDD)
 - ☐ P2-F27-T06 — `internal/repomap/repomap_tool.go`: repomap tool (TDD)
 - ☐ P2-F27-T07 — /aider slash command (TDD)
 - ☐ P2-F27-T08 — `main.go` wiring + integration tests
 - ☐ P2-F27-T09 — Challenge harness 6 phases + close-out + push 4 remotes
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Task 1: Bootstrap F27

Advance PROGRESS.md to “F27 in flight”. Update CONTINUATION.md F27 mid-flight section. Append evidence header.

Task 2: voice/types.go + recorder.go

- VoiceRecorder with Start(outputPath string), Stop(), IsRecording() bool
- Shell: arecord -f cd -t wav <path> (Linux) with sox fallback
- Subprocess management via os/exec, signal-based graceful stop
- WAV validation: check RIFF header, non-zero data chunk
- Tests: mock recorder for unit tests, real recorder gated (SKIP-OK)

Task 3: voice/transcriber.go

- VoiceTranscriber with Transcribe(audioPath string) (string, error)
- Primary: OpenAI Whisper API (POST /v1/audio/transcriptions)
- Fallback: whisper.cpp CLI (whisper-cli -m <model> -f <wav>)
- Env vars: OPENAI_API_KEY for API, HELIXCODE_WHISPER_MODEL for local model path
- Tests: mock HTTP server for API, mock CLI for fallback

Task 4: voice_tools.go

- voice_start: starts recording to temp WAV
- voice_stop: stops recording, returns path + duration
- voice_transcribe: transcribes stopped recording, returns text
- Category: voice. RequiresApproval: transcribe→LevelEdit, start/stop→LevelReadOnly

Task 5: repomap/mapper.go + cache.go

- Extend existing internal/repomap/ package
- Tree-sitter parser: for each source file, parse AST, extract function/class/import nodes
- Map struct: RepoMap{Files []FileMap} where FileMap{Path, Functions, Classes, Imports}
- Cache: Cache{mu, entries map[string]*RepoMap} keyed by git rev-parse HEAD
- Tests with real tempdir repos containing Go/Python/JS files

Task 6: repomap_tool.go

- repomap tool: scans project dir, returns structured map as JSON
- Category: mapping. RequiresApproval: LevelReadOnly

Task 7: /aider slash

- /aider voice [start|stop|transcribe] — voice control
- /aider repomap [generate|show|clear] — repo-map control
- Aliases: /aider, /ai

Task 8: main.go wiring

Register 4 tools + /aider slash in main.go startup.

Task 9: Challenge harness

6 phases: - A: Voice record (gated on microphone) → WAV > 44 bytes - B: Voice transcribe (gated on API key/model) → non-empty text - C: Repo-map on tempdir with Go files → finds functions - D: Repo-map cache → same commit returns cached, new commit invalidates - E: Hybrid fallback → API absent exercises whisper.cpp path - F: /aider slash subcommands → correct routing

Plan written. Execute via TDD starting with T01.